

Autonomous Guided Vehicle Project

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Abstract

This report documents the design and development of an Autonomous Guided Vehicle (AGV) system in a simulated warehouse environment. It describes the current controller, earlier controller iterations, the logging infrastructure and the web application used to inspect simulation results. Replace this text with a concise summary of the final system, its main design choices and the outcomes of the evaluation.

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1 Current Controller

This section describes the current AGV controller implemented for the Webots warehouse simulation. The controller integrates perception, localization, planning and low-level motion control to move each vehicle through the environment while accounting for configured goals and dynamic obstacles.

1.1 Architecture

Describe the controller as a sequence of cooperating components. Start from the information received from Webots and explain how sensor data is transformed into a state estimate, a route and finally motor commands. Refer to implementation details only when they clarify an engineering choice or a relevant limitation.

1.2 Planning and Control

Explain the high-level and low-level planning responsibilities, including how the controller selects waypoints, reacts to blocked paths and completes a goal. Use equations only when they make an algorithm or a measurable quantity clearer; otherwise, prefer a precise prose explanation.

1.3 Observed Behaviour

Summarize the behaviour observed during representative simulations. State what worked reliably, identify situations that required special handling and connect each limitation to its likely technical cause.

2 Previous Controller Iterations

The current controller was preceded by earlier implementations that helped establish the project requirements and expose practical limitations. This section records their main design choices so that the evolution of the system remains traceable.

2.1 Initial Approach

Describe the responsibilities and operating assumptions of the first controller. Focus on the motivation for the approach, the parts that were effective and the conditions under which it no longer scaled to the warehouse scenario.

2.2 Lessons Learned

Compare the earlier approach with the current controller in terms of robustness, maintainability and behaviour around dynamic obstacles. Present limitations as engineering observations rather than as unsupported claims, and explain how they informed subsequent design decisions.

3 Logging Infrastructure

The logging subsystem provides a persistent record of simulations, events and telemetry generated by the AGVs. Its role is to make the execution of a simulation inspectable after the fact and to provide the data consumed by monitoring tools.

3.1 Service and Data Flow

Explain how controllers communicate with the logging service, how the service stores data and which entities are represented in the database. Clarify the lifecycle of a simulation record and the information associated with individual events.

3.2 Reliability Considerations

Document the connection assumptions, retry behaviour and failure modes that affect the controller. Explain why these choices were made and how they balance timely logging with safe simulation execution.

3.3 Use in Evaluation

Describe how stored telemetry and events support debugging, performance analysis and the comparison of simulation runs. Include concrete measurements or examples when available.

4 Web Application

The web application provides an operational view of the AGV simulations. It combines the data exposed by the logging pipeline with a visual representation of the warehouse, allowing users to inspect vehicle status, configured goals and the progression of a run.

4.1 Interface Design

Describe the principal views of the application and the information each view presents. Explain how the interface supports the workflow of monitoring a live simulation and reviewing a completed one.

4.2 Data Integration

Explain how the web application obtains and transforms simulation data before rendering it. Mention refresh behaviour, data formats and any consistency constraints that affect what users see.

4.3 Practical Value and Limitations

Summarize the value of the dashboard for development and demonstrations. Then state the current limitations, such as dependence on the logging service or any incomplete visualizations, and identify the most useful future improvements.